

5.3 UBISWORLD and UBISONTOLOGY

Human-computer interaction in ubiquitous computing needs a uniform virtual world model in order to simulate, represent and compare research issues. The initial idea behind UbiWorld is the extension of the Blocks World, see [Slaney and Thiébaux, 2001], and the extension of the context toolkit, see [Salber et al., 1999a] to the special needs of situated interaction in ubiquitous computing with user modeling and privacy. The prefix *Ubi* abbreviates the term *Ubiquitous*. The postfix *World* indicates, that our approach tries to be very broad and it refers to the blocks world. UbiWorld can be seen as a collection of concepts and models for location and time, for interaction and situation that are all prepared for ontological representation and data collection. UbiWorld has so far been described in [Heckmann, 2003a], [Heckmann, 2003c], [Stahl and Heckmann, 2004b], [Kruppa et al., 2005] and [Compiol, 2005]. With this virtual intelligent environment we are able to run experiments for ubiquitous computing. The next subsection shows the full inter-relationship between the real world and UbiWorld.

From Real World to UbiWorld

UbiWorld can be used to represent some parts of the real world like an office, a shop, a museum or an airport. It represents persons, objects, locations as well as times, events and their properties and features. Apart from the representational function, UbiWorld can be used for simulation, inspection and control, but the question “*What parts of the real world should be represented?*” forms the first step in the diagram of figure 5.16.

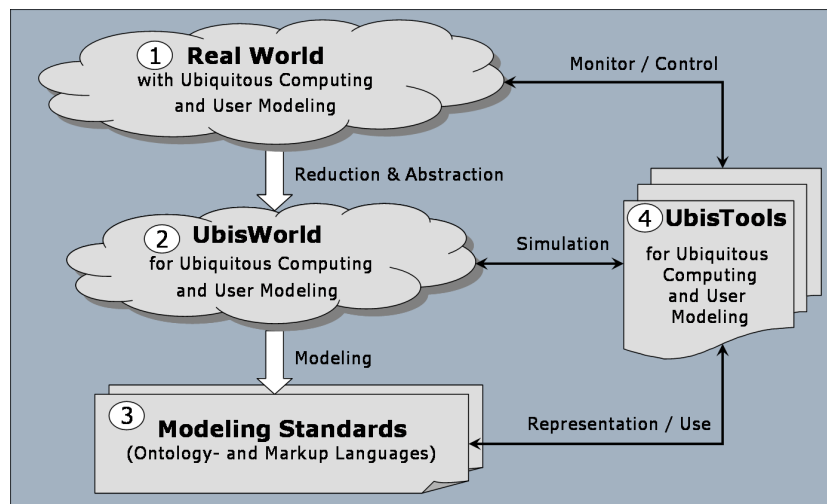


Figure 5.16: Conceptual overview: from real world with ubiquitous computing and user modeling (1) to UbiWorld (2) with modeling standards (3) and specialized tools (4)

Cloud number (1) in figure 5.16 stands for the points of interest of the real world with ubiquitous computing and user modeling. UbiWorld, displayed as cloud number (2), is real-

ized by reduction and abstraction from the real world. The modeling standards like markup-languages and ontology definitions are derived from UbiWorld abstractions.

It is interesting about the role of UbiTools (4) for ubiquitous computing and user modeling, that on one hand operate on the level of UbiWorld and on the other hand operate on the level of the real world.

A small example to illustrate the interrelation between a simulated world and a real world: *Imagine that there is a room in the real world with two doors, two light switches at each door and one light at the ceiling. All these elements will be represented in an abstract manner in the corresponding UbiWorld model. The UbiTools could simulate the light-on light-off behavior of the real room, such that if the virtual light switch gets pressed the virtual light in UbiWorld shines, independently from the real world. Secondly, the UbiWorld room could be used to monitor the real room in such a way that every time when the real switch has been used, the virtual light shows the status “shining”. As a third possibility, the UbiWorld room could be used to control the real world room for example by turning the real light on or off, every time when the virtual light switch is used. What follows - from this example - is that the ontology engineering of the UbiWorld ontology should be independent from the later task of representation, simulation, monitoring or even control of the real world.*

A second important point, before one can start with modeling the UbiWorld ontology, is that we need at least a rough idea about the interaction model for ubiquitous computing with integrated user modeling and privacy. Figure 5.17 shows a simplified human-computer interaction model, where the context, the location, the time and the sensing & computation are schematically wrapped around the user with his or her user model and the system with its resource model.

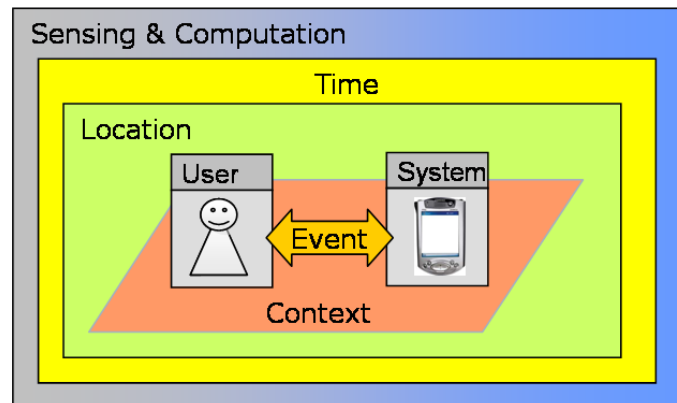


Figure 5.17: Simplified human-computer interaction model for ubiquitous computing with user modeling and privacy

The *User* icon and the *System* icon are also meant to represent user groups and collections of interacting systems. In section 2.2.4 a more complex interaction model has been discussed.

The colors of the simplified human-computer interaction model as shown in figure 5.17 and in figure B.1 on page 230 are used to classify the corresponding partial ontologies. Since the semantic web ontologies support and encourage the integration of several ontologies, the design decision for UbiWorld was to develop specialized partial ontologies, rather than trying to realize one ontology for all aspects. Figure 5.18 shows in one diagram the elements of the UbiWorld concept, which consists of classes and predicates, of individuals, and of relations. In UbiWorld the classes and predicates are defined in six additive ontologies, namely the *physical ontology*, the *spatial ontology*, the *temporal ontology*, the *activity ontology*, the *situation ontology* with situation describing dimensions that also cover the general user model ontology GUMO and finally the *inference ontology* which models the computational and intelligent behavior in ubiquitous computing environments.

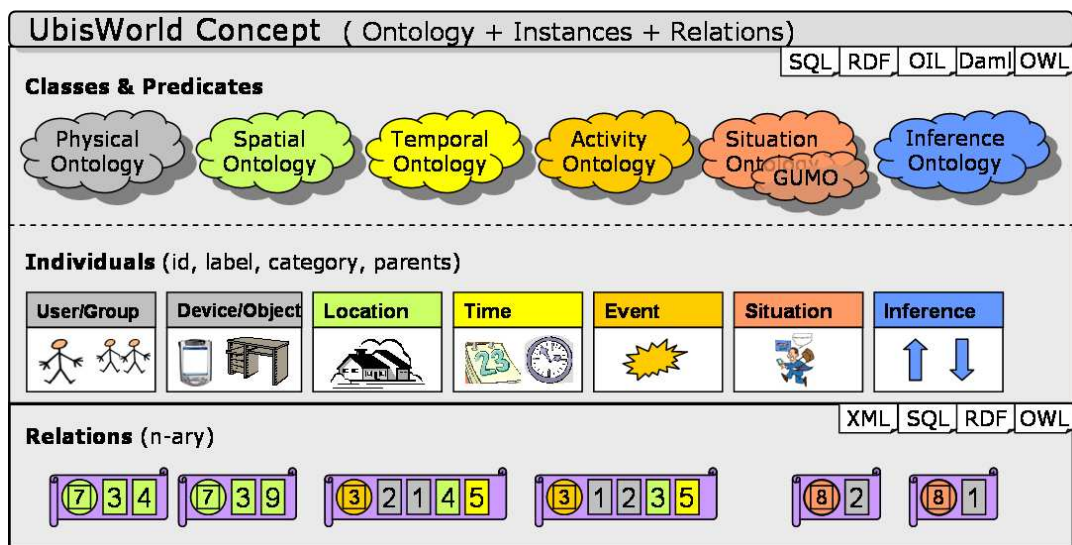


Figure 5.18: *UbiWorld = Ontology + Instances + Relations*

Figure 5.18 indicates, that all ontologies are available in several ontology representation languages, among which are RDF, OIL, Daml and OWL. This verbosity allows a maximal reusability for different systems. All these different instantiations of the ontologies are realized by a generation and translation tool. Some parts of the ontologies can be found in the appendix B.2. In section 8.2.2 an ontology editor is described, that allows the online inspection, adding and editing of the ontological elements. Also important to mention is the fact, that the design decision for UbiWorld was to work with n-ary relations rather than binary relations only, even though they are not yet directly realized within the new semantic web approach. N-ary relations are especially important for the integration of SITUATIONAL-STATEMENTS that can contain up to 25 positions per relation. Figure 7.7 on page 147 shows the integration of SITUATIONALSTATEMENTS into the rest of the ontology. In the following subsections, the concepts behind the partial ontologies are discussed.

5.3.1 Physical Ontology: Users, Groups, Devices and Objects



The physical ontology introduces physical objects, that are especially important under the given constraint of ubiquitous computing and user modeling. With physical elements we think of persons, devices, objects, furniture, goods, food and so on, everything that can play an important role in so-called intelligent environments. The ontology model divides between the three basic elements Being, Thing and System, and element groupings like UserGroup, SystemGroup and MixedGroup. Every basic element carries a default class icon. Important categories for the ubicomp¹⁴ domain are User, Device, Furniture and Vehicle. All other graspable objects will be defined under the category OtherObject. Of course this categorization is arbitrary, however, it is of minor importance. More important is the correct identification and the inheritance of class properties to the instances. Figure 5.19 shows the partial top-level ontology of the physical ontology with the focus on the DeviceElements.

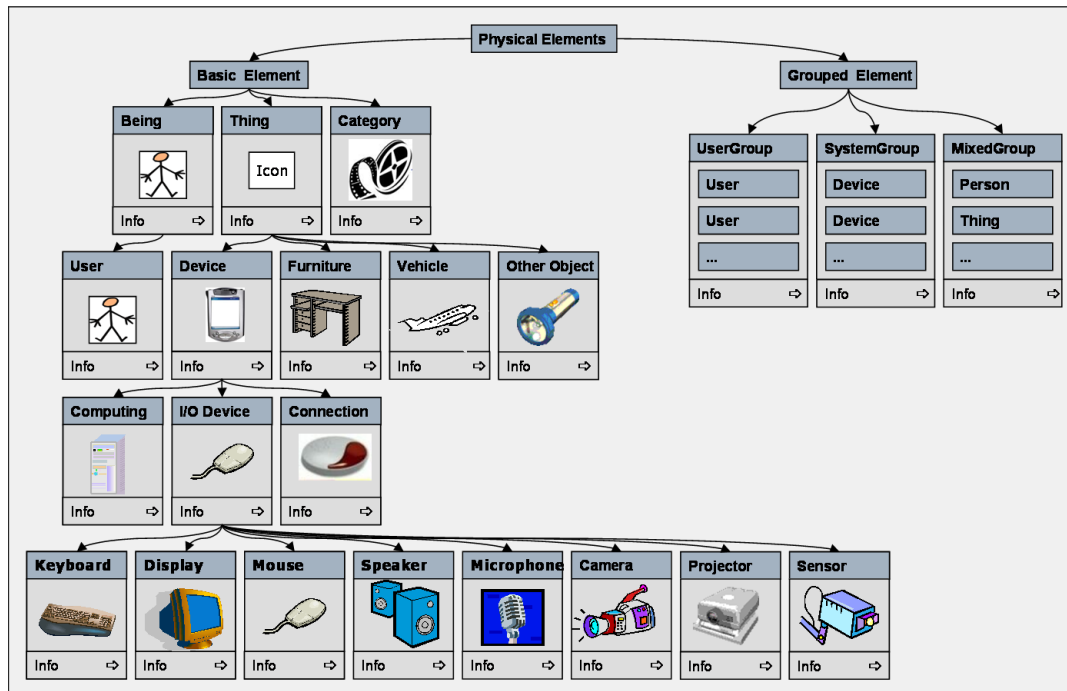


Figure 5.19: Some physical elements in the so-called physical ontology with the focus on device elements

Input/output device elements are for example: Keyboard, Display, Mouse, Speaker, Microphone, Camera, Projector, IR Bark, ID TAG, Notebook, Mobile Phone and so on. They are important to model the instrumentation in the environment.

¹⁴ubicomp = abbreviation for ubiquitous computing

Each element in the ontology has its own web page as a virtual representative. Figure 5.20 shows the class `SmartShopObjects` as sub-class of the class `Thing`. This illustrates the close connection between the real world scenarios and the formal ontology. Its ancestor elements and its child elements are presented on this web page as hyperlinks. The concept behind the interlinked elements in the UbiWorld ontologies is a mixture between *frames with slots* and *semantic nets*.

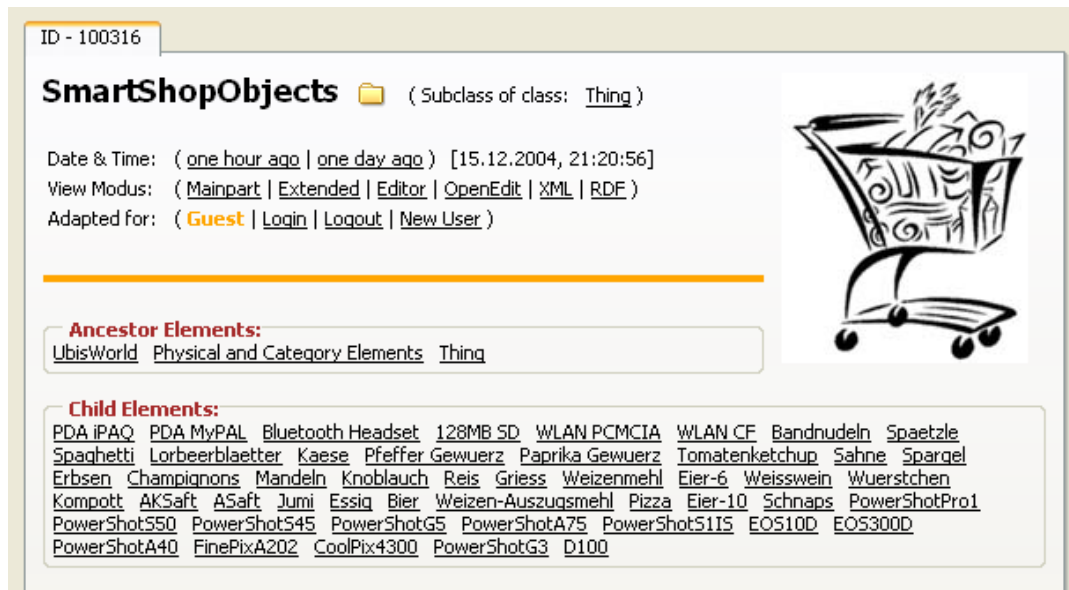


Figure 5.20: The class `SmartShopObjects` (in the UbiWorld Element Browser) is a container for several physical elements that are used in the smart-shop scenario

Since our ontology allows multiple inheritance, an object like the *Canon PowerShot S50* can be the child in the device hierarchy, but at the same time be the child of the `SmartShopObjects`. Figure 5.21 shows the virtual representative of an instance of this camera. All products in the `SmartShopObjects` class have at least the product information *vendor* and *price*. No problem arises if we describe it as a product-type or product-instance, since both are equally handled as entities within UbiWorld. Further information about physical objects that is of importance for ubiquitous computing is the degree of mobility. Objects can be *mobile*, *movable* or *fixed*. Some objects may be small, but not necessarily mobile. A mobile object is for example a PDA. A movable object for example is a desktop PC. It is not meant to be moved regularly. The degree of mobility has an influence on the default change model. Furniture like a table, a shelf, a bed, or a chair are also defined as being movable. If no statement is mentioned about a movable or fixed object, it is assumed that its location has not changed. See section 4.3.4 on page 67 for a discussion of the frame problem.

Another interesting property of objects and devices in the PhysicalOntology, which is especially important for ubiquitous computing, is the distinction between the *public* and



Figure 5.21: The virtual representative of the Canon Powershop S50

private status. The SPECTER concept as described in [Kleinbauer et al., 2003] is a private - mobile device, while the large screen in the instrumented environment as described in [Stahl et al., 2004] is a public - fixed device. In [Kruppa and Krüger, 2003] the importance of public and private devices is discussed.

5.3.2 Spatial Ontology: Location, Topology, Orientation



Physical objects and devices are spatially arranged while the users move and interact in mobile and ubiquitous computing with their everyday environments. Services are location-aware while local awareness plays an important role. Thus a profound investigation of spatial concepts and models has to be undertaken, see [Stahl and Heckmann, 2004b] for an extended summary of this research. The first spatial concept analyzed here is the one of spatial granulation levels. As stated above, mobile and ubiquitous computing shifts the human-computer interaction from the desktop in one room into the halls of an airport or shops. In the travel domain, whole countries or continents can be the point of interest for description. Figure 5.22 shows nine significant spatial granularity levels from *Country* over *City* and *Building* to *Room* and even to the *Object Level* that can be identified in connection with the research about human interaction.

The distinction between *handling of objects*, *indoor moving within rooms* and *indoor moving between different rooms*, *outdoor moving*, *car&bus driving* as well as *airplane&train traveling* is interesting, since different techniques for the detection of the user's location are applied and different ways of adaptation are introduced according to their corresponding

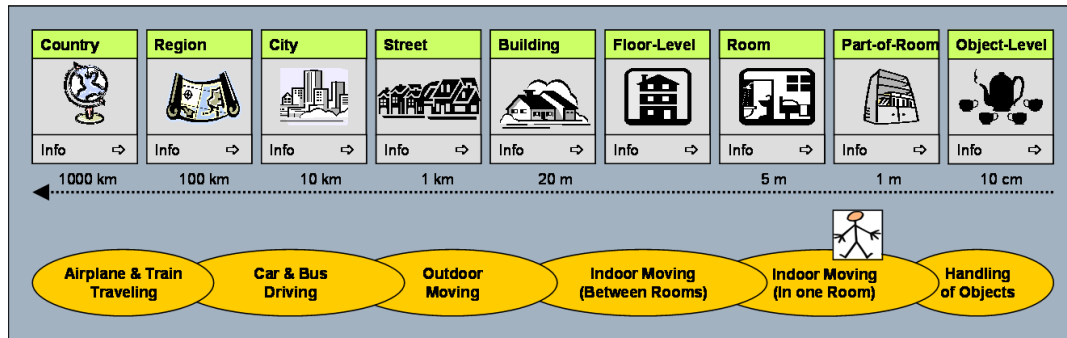


Figure 5.22: Spatial granulation levels from “country” over “building” to “object level”, together with corresponding human mobility interactions from “airplane and train traveling” to “handling objects”

spatial granulation levels. Instances of qualitative spatial elements are rooms, buildings, cities, streets, and all can be mapped to these spatial granularity levels. Furthermore all instances of spatial and physical elements can be spatially arranged and different relations can hold between them. An example visualization of the spatial relation “is-nested-in” of physical and spatial elements is presented in figure 5.23.

This partial topology model puts the focus on the offices of the *WW-floor*¹⁵ in *building E1 1* at the Saarland University. Even persons and their mobile objects that they carry along are displayed in the topology. This tree visualization enables a fold- and unfold-functionality of spatial locations. Thus the UBISLOCATIONMONITOR, that displays the context of locations, can be called for every qualitative location instance by browsing the location tree and clicking to any node. Figure 5.24 shows the UBISLOCATIONMONITOR for *Room 124* which is an instance of the class *Room*. The displayed information is adapted to the person that is currently logged in, in this case an anonymous guest.

The attribute *Location Path* shows the parents of room 124 and their is-nested-in relations. The question that now can be handled situation-aware by this information is *Where is Rainer?* The adapted answer in this spatial model could be *Room 124* for example for a call within the floor, it could be in *building E1 1* for a request within the university, or it could be *Saarbrücken* in some cases, or even *Germany*. To enable such adaptive answers automatically without a spatial inference system, the UBISLOCATIONMONITOR offers the whole location path in form of a web service. The attribute *Ancestor Elements* is related to the ontology and classifies the object as spatial element. The attribute *Physical Elements* in this *Location* says that *Rainer* is currently in this location. Furthermore, the *Temperature* is said to be high, while the *Noise Level* is said to be low. Thus spatial elements can have properties in form of *SITUATIONALSTATEMENTS* just like users or physical objects, however most of the predicates differ.

¹⁵This location model can be found at <http://www.u2m.org/ubisworld.htm>. It has partly been modeled by Holger Schultheis in [Morsing and Schultheis, 2004]

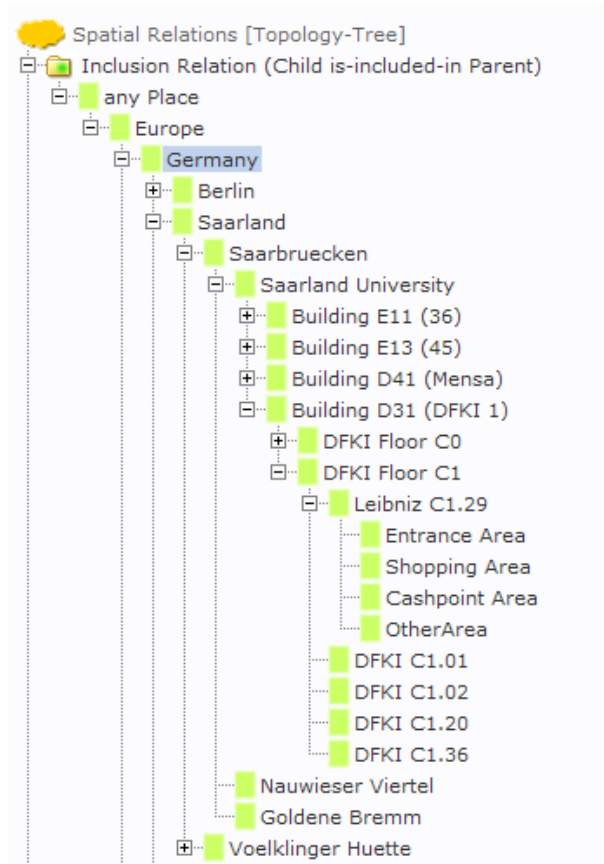


Figure 5.23: Example visualization of the spatial relation “is-nested-in” in UbisWorld

An important problem that has to be solved and integrated in the spatial ontology is the fundamental structural difference between qualitative or symbolic location models (as described so far) and quantitative or geometric location models with coordinates (as supplied by a GPS¹⁶ receiver) in two, two and a half or three dimensions. In [Stahl and Heckmann, 2004b] a new hybrid location modeling has been introduced with the intention to model location in order to realize location aware applications with a focus on situated user interaction and pedestrian navigation. The tool YAMAMOTO¹⁷ has been developed for the modeling of hierarchical geometrical maps. Figure 5.25 shows the map reference functionality that combines the symbolic topology on the right hand side with the corresponding coordinates on different maps on the left hand side.

The tool YAMAMOTO, that is described in section 8.12 on page 166, represents the geographical coordinates of real-world places in different granularity as resources in the world wide web. The geometrical model is joined with the symbolic model by uni-

¹⁶GPS = Global Positioning Service

¹⁷YAMAMOTO homepage: <http://w5.cs.uni-sb.de/~stahl/yamamoto/>

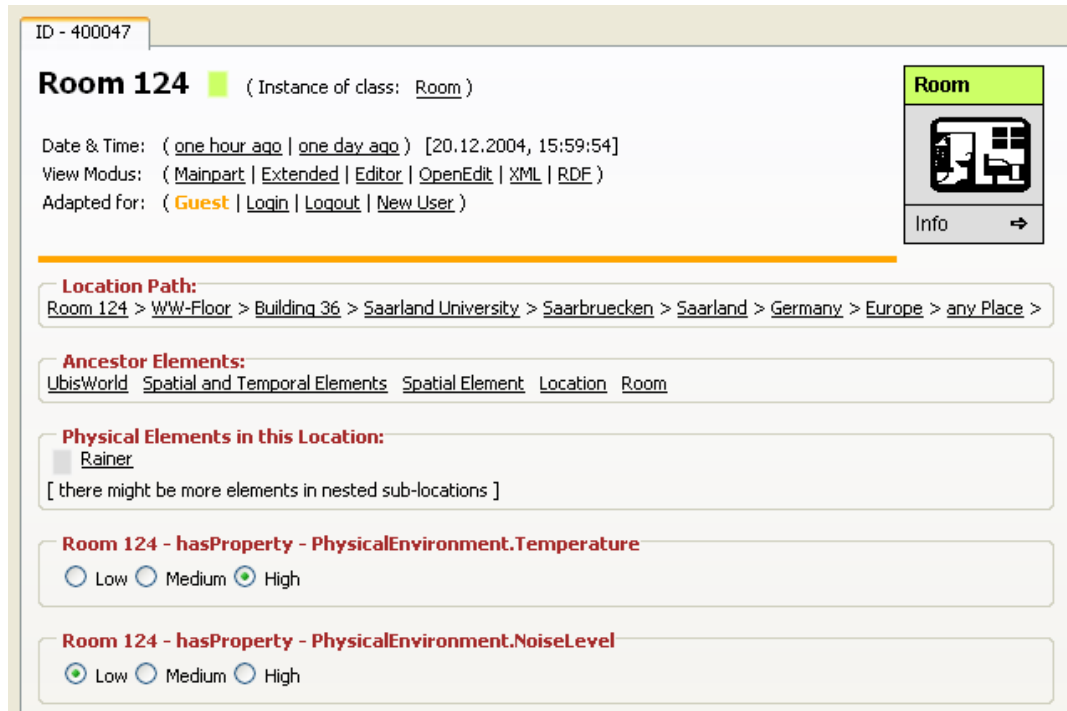


Figure 5.24: UBISLOCATIONMONITOR for Room 124

form resource identifiers (URIs) as introduced in section 4.5 on page 74. The mobile systems in our scenarios locate themselves outside buildings using a GPS receiver, while indoors the intelligent environment provides localization based on an infrared beacon and RFID tag infrastructure, see [Stahl et al., 2005]. Symbolic and geometric location models are discussed in [Leonhardt, 1998] where a combined model is proposed. In [Jiang and Steenkiste, 2002] the aura location identifier ALI is introduced. It is based on hierarchical subspaces to realize a space service to handle spatial queries with a relational database. [Dürr and Rothermel, 2003] suggest a lattice instead of a tree structure to model hierarchical symbolic locations in order to express complex containment relationships between rooms to wings and floors of a building. In [Baus et al., 2005] an up-to-date survey of map-based mobile guides is presented. The NEXUS project¹⁸ provides a platform for location-based services. It relies on an augmented world model, that represents real world objects as well as virtual objects. The world model is hierarchically decomposed into area models, which are provided by a spatial model server infrastructure. The positions of mobile objects are stored in location servers, which are designed to handle spatial queries on a large scale. The complete concepts of the UbisWorld spatial ontology, that are based on the related work discussed above, are shown in figure 5.26.

Spatial elements are divided into the two basic concepts of Location and

¹⁸NEXUS, Sonderforschungsbereich SFB 627, <http://www.nexus.uni-stuttgart.de>

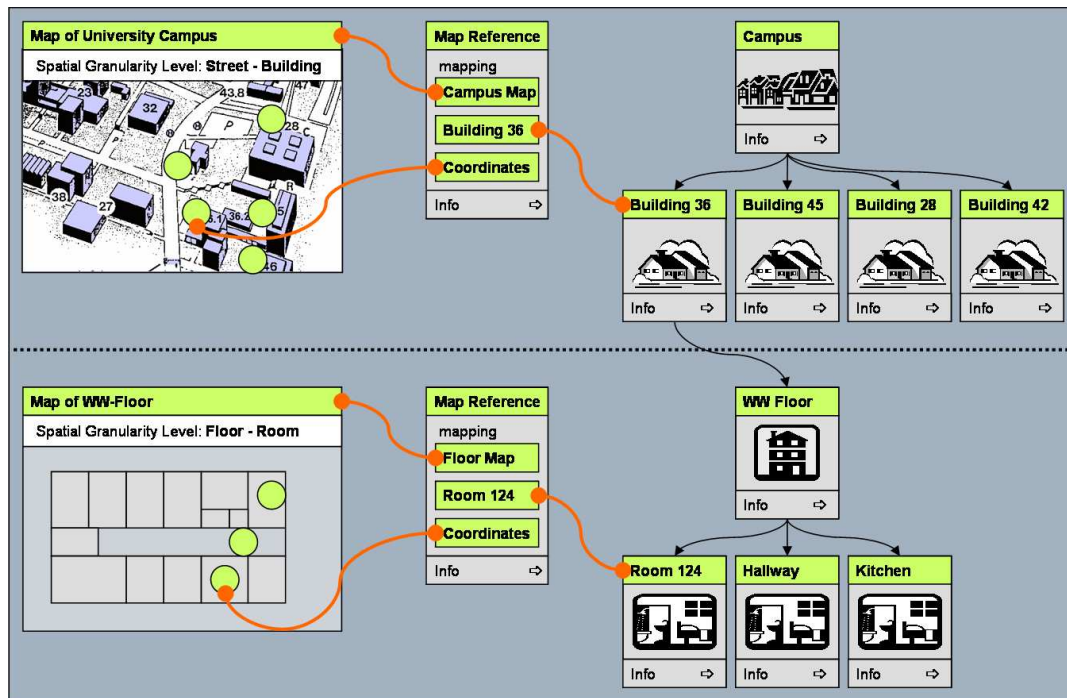


Figure 5.25: Map referencing and the hybrid location model

SpatialConstraints. The first one can be a Place with the corresponding spatial granularity level, an Area where a spatial extension is given, an Entity Location, that defines the location next to a physical object, or a Relative Location that defines locations like “between A and B”. The second one, the spatial constraints, are Connection, that relates two locations by a connection, Nesting that relates two locations by the is-nested-in relation, Map Reference as discussed above, Constraint that allows for arbitrary spatial relations, and finally Orientation. This expressivity is helpful for modeling and inferencing in ubiquitous human-computer interaction.

5.3.3 Temporal Ontology: Point of Time, Interval, Temporal Constraints



The third partial ontology in UbiWorld is the temporal ontology. A clear model of time and time-intervals is essential, since most statements are related over the temporal dimension. The user interface design decisions allow for historical views of the user- and situation models. The real life offers verbose descriptions of time. Systems have to be synchronized, sometime the temporal knowledge is approximate or statements about the future have to estimate durations. In analogy to the spatial concept, a hybrid temporal concept has been developed. Figure 5.27 shows the concept of temporal granularity and the defining elements of a timestamp.

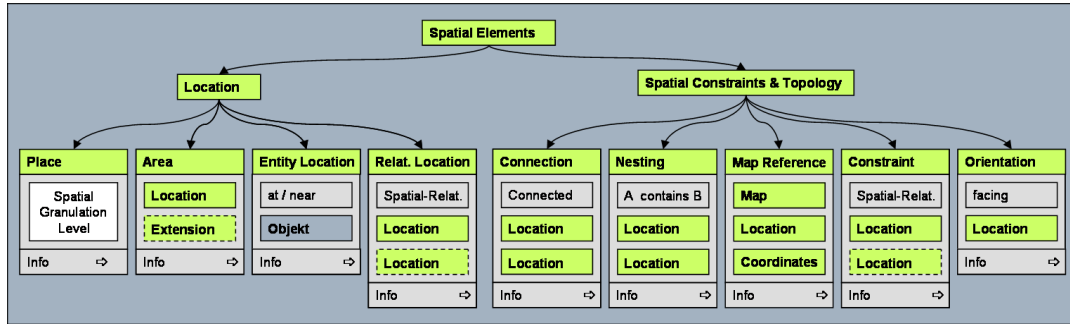


Figure 5.26: Spatial elements that form the basic spatial concepts for UbiWorld

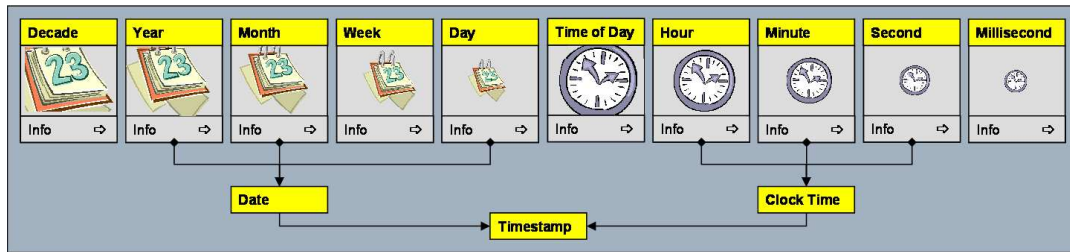


Figure 5.27: Temporal granularity and timestamp

The dimension of time is dense in such a way that every event has a temporal extension. Thus an interval seems to be the right concept to model time related issues, however by introducing the temporal granularity, one can also work with points of time. The duration aspect in the temporal domain corresponds to the concept of distance in the spatial domain. The date and clock time could be compared to the address in the spatial domain. The analogy between the temporal model and the spatial model can be seen if figure 5.26 and figure 5.28 are compared. If a time is asked in a question, it is not natural to simply state the full timestamp as answer, but the temporal granularity has to be adapted to the situation. For example to the question *When did something happen?* several temporal granularity levels could be the right one, like *in the 60s*, or *in 1969*, or *last Friday* or even a complex expression like *between event A and event B*. The first one describes the point of time by a whole decade, the second one by a year. The event itself could have happen within minutes, but the accuracy of the statement varies according so the situation. Thus the expressivity should be allowed. The temporal aspects within SITUATIONALSTATEMENTS can be found in section 4.2.

Figure 5.28 shows the temporal elements and divides between Time and TemporalConstraints&Chronology. The first one can be a Point of Time with its corresponding temporal granularity, or a Time Interval with a start time and either the duration or the end time. Time by Event defines the time by referring to an event. Time by Relation defines the time by unary or binary temporal relations. The second one, the temporal constraints and chronology, allow a Partial Order of the time by the

before relation. They allow temporal Nesting or any other temporal Constraint like after or overlap. The currently implemented list of temporal elements as shown on the

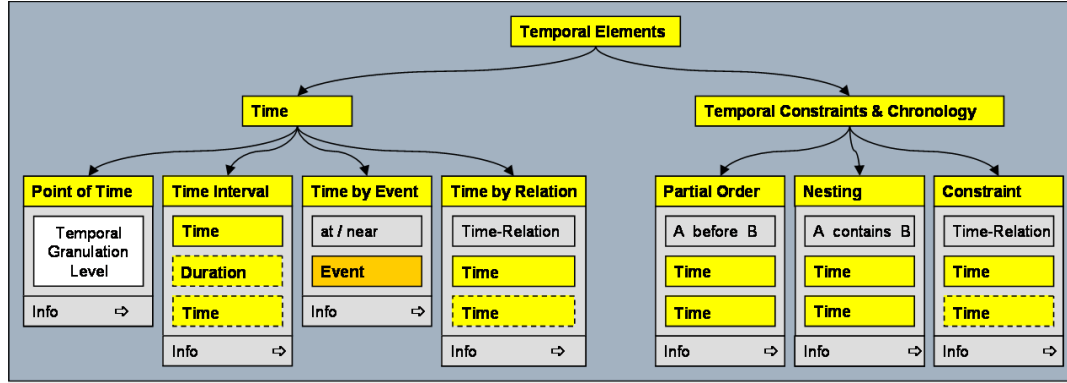


Figure 5.28: Temporal elements: point of time, intervals, temporal constraints

UbiOntologyBrowser can be found in the appendix B.2.3 on page 236.

5.3.4 Activity Ontology: Changing the World



With physical objects, spatial elements and temporal elements, as introduced in the three partial ontologies above, one can already describe complex but static worlds. Now, the activity elements describe the changes in the world and the most prominent one is *change of location*. Some events take place in point-of-times, others take place over time intervals. Four concrete example activities from the shopping domain are shown in figure 5.29. Such activities are described in SITUATIONALSTATEMENTS, such that the agents and time constraints and other important additional information can be stated or left underspecified.

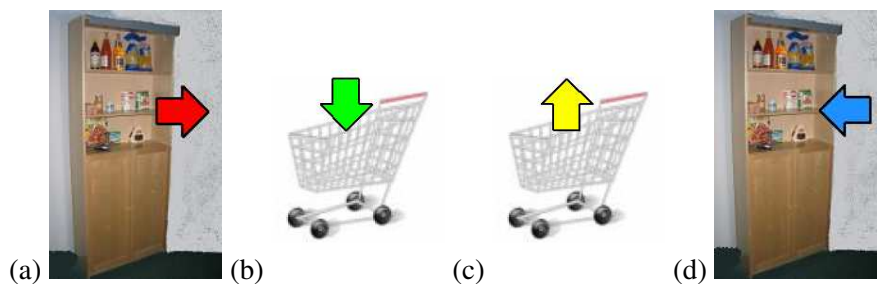


Figure 5.29: Some examples of activities in the shopping domain: (a) shelf-out, (b) cart-in, (c) cart-out, (d) shelf-in

Thus, the blocks-world that can be formed with elements from the UbiWorld ontology is not activity-centered or relation-centered. All types of elements (physical, spatial, temporal, activity, situation, inference) are equally important and arranged around SITUATIONAL-

STATEMENTS, that form the center of the whole approach. This view is established on the idea of *realism* in situation semantics, see [Barwise and Perry, 1983], where basic properties and relations are taken to be real objects and not sets of n-tuples or functions. Figure 5.30 shows some classes of activities in the view of the UBISONTOLOGYBROWSER. Move, Take, Give, Put and Change are the most basic ones. Administration activities are for example Login or Logout.

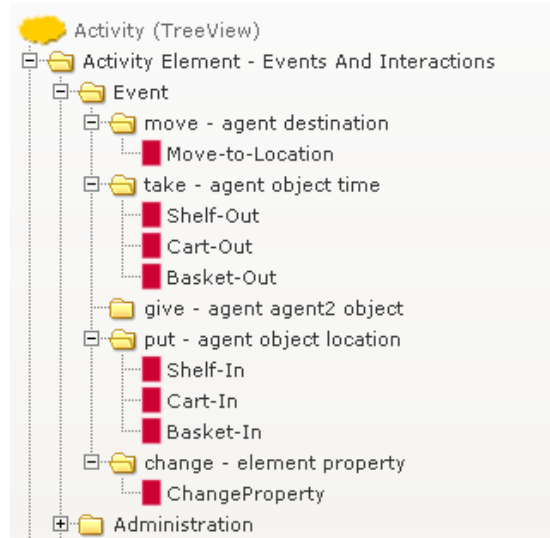


Figure 5.30: Some activity classes as shown in the *UbisOntologyBrowser*

This activity ontology finds its purpose in the software engineering and not in the research of proper classification and definition of concepts. We simply need resource identifiers to refer to the special activities and events, and the interoperability is realized by pointing to the same class or instance. Later on, in further research, either a profound ontological definition of the activity concepts will be needed, or attempts for further integration of the SUMO/MILO ontologies have to be undertaken. Figure 5.31 shows the conceptual diagram of the location events Translation, Rotation and Pointing. To represent an event or interaction, there are also slots with physical elements, spatial elements and temporal elements needed.

Not mentioned - so far - is the conceptual link to further information [Info $\Rightarrow$] as shown in the figures 5.31, 5.28, 5.27, and 5.19 for example, that describes the element or interaction in detail. This additional link allows to state contextual attributes and situational descriptions to every element or compound element structure in the conceptual model. The connection between the conceptual model and the UBISONTOLOGY is shown in the following subsection.

Figure 5.32 shows the graphical user interface of a basic activity editor to enter *move-to-location* events, and the *Actual List of Activities* monitor. Both functionalities need to be extended. The monitor displays the events that have been entered by the editor, but also the events that have been measured and inferred by the systems. Specialized monitors for the

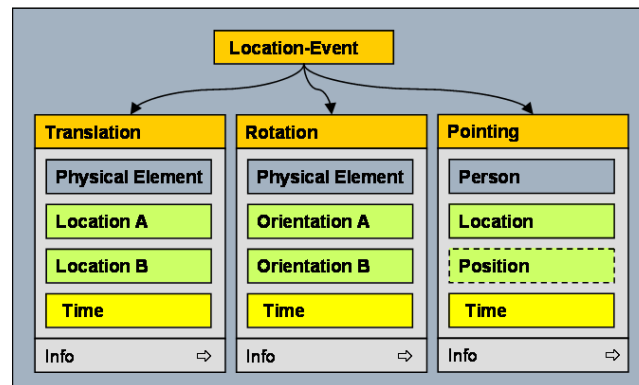


Figure 5.31: Location Events in the Conceptual Model

shopping domain and the pedestrian navigation domain are described in section 9.6.

5.3.5 Situation Ontology: Describing the World



Attributes, parameters or properties about users, systems, locations or activities, are collected in the *Situation Ontology*. A situational parameter for a location could for example be the Noise Level, the Weather conditions, or the available Light. A situational parameter for a person could for example be his/her Blood Pressure, his/her Cognitive Load or his/her Interests. Because of the importance and relevance for the research of ubiquitous user modeling, the situational parameters that concern users, have been analyzed and defined in more detail in section 5.2, starting from page 85. Hence, the defined general user model ontology GUMO can be considered as being part of the Situation Ontology. A situational parameter about a device or system could for example be its remaining Battery Power or its Screen Size. Figure 5.33(a) shows some situation elements from the context information subtree in the UbiOntologyBrowser.

The subclass *Social Environment* points towards the contextual group and team interactions. Examples for attributes from the *Physical Environment* are given in the screen shot in figure 5.24, and examples for attributes from the *Product Information* are given in the screen shot in figure 5.21.

A further distinction in the situation ontology can be made, between low level sensory data and higher level inferred data. Even though there is no clear possibility to distinguish between the different higher levels of inference¹⁹, the low level sensor

¹⁹For example, the Age of a person can easily be estimated by the ages of elder and younger sisters and brothers. It could also be calculated in a complex inference process through speech analysis, as done by Müller in [Müller and Wittig, 2003], on the other hand, it could be analyzed by complex image processing from a photo on a birthday card. Thus the situational parameter itself is independent from the number of inference processes, that produce the value for the situational parameter.

Introduction SemanticWeb Location **Activity** Inference Query

Enter an Activity to Ubi's World:

Move-to-Location

Subject:

Location:

Actual List of Activities

- [475] 29.12.2004, 11:18:00 ■ **Move-to-Location** ([White Chocolate](#) , [AirportBerlin](#))
- [474] 29.12.2004, 11:16:20 ■ **user login** ([Guest](#) , [UbiWorld](#))
- [473] 27.12.2004, 07:24:26 ■ **user login** ([Guest](#) , [UbiWorld](#))
- [472] 22.12.2004, 10:23:26 ■ **user login** ([cristinagena](#) , [UbiWorld](#))
- [471] 21.12.2004, 06:55:56 ■ **user login** ([niuty](#) , [UbiWorld](#))
- [470] 16.12.2004, 11:48:32 ■ **Move-to-Location** ([Christoph](#) , [Office106](#))
- [469] 00.00.0000, 00:00:00 ■ **ChangeProperty** ([Christoph](#) , [Sound](#))

Figure 5.32: Activity Editor and the UbiActivityMonitor

data can roughly be classified by the mass of uninterpreted data, being sent on a regular, frequent basis, from a special sensor device. Figure 5.33(b) shows some low level sensor data elements from the Speech Parameters, that were implemented in [Wasinger et al., 2003a], Biometrical Sensor Data, that were implemented in [Brandherm and Schmitz, 2004], and Typing Behavior, that were implemented in [Lindmark and Heckmann, 2000].

Possible values for the Speaking Style are *formal*, *informal* and *childish*. The question that arises is: *Should we add data types and ranges to the ontology or should we define them separately?* What turned out to be treatable in practice was modeling the distinct value sets within the ontology but to define the dense ranges as data types outside the ontology. Figure 5.34 shows several distinct value sets. Each data type receives an identifier and by that a referable URI. Furthermore, each possible value receives an identifier and can thus be semantically described and interpreted by the means of the ontology. Most of the situation parameters are combined with a default data type, that could be overwritten if needed. For example the Temperature in a room could be measured in degrees *Fahrenheit* or in degrees *Celcius*, or its range could be the set { *cold*, *normal*, *warm*, *hot* }. As stated in the chapter about SITUATIONALSTATEMENTS, the predicate is decoupled from the used range. On the one hand, a data type like `PoorGoodPerfect.640040` can be used for many attributes, while on the other hand the data type `NoiseLevelType.640160` seems to be only useful for the physical environment parameter `NoiseLevel.820010`. However, this data type could for example also be used to classify the happiness of a baby: { *quiet*,

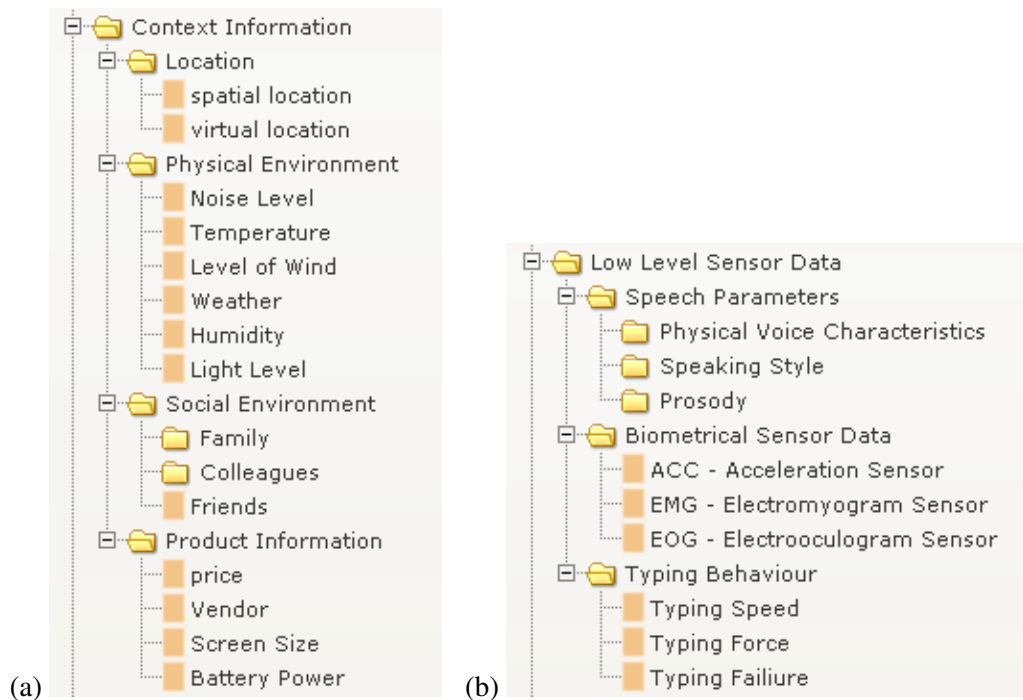


Figure 5.33: (a) Situation ontology elements for context information (b) Situation ontology elements, classified as low level sensor data

normal, noisy, loud, extremely loud }.

5.3.6 Inference Ontology: Computing the World



Finally the power of computing and intelligent behavior enters the Ubi-World by the *Inference Ontology*. Inference elements define smart rules or proactive inference processes within intelligent instrumented environments. In ubiquitous computing and ambient intelligence, several scenarios have been proposed, among which the `home scenario` and the `office scenario` are the most prominent ones. In the *Georgia Tech's Aware Home Research Initiative*²⁰, see [Kidd et al., 1999], the house as shown in figure 5.35(a) is aware of its occupants whereabouts and activities. They analyze the question: *If we build such a home, how can it provide services to its residents that enhance their quality of life or help them to maintain independence as they age?* The Aware Home Research Initiative is an interdisciplinary research endeavor at addressing the fundamental technical, design, and social challenges presented by such questions and experimenting with the future of domestic technologies. The Microsoft Research project *EasyLiving*²¹ is developing a prototype architecture and technologies for building intelligent environments. A survey of research on context

²⁰Georgia Tech's Aware Home Research Initiative: <http://www.cc.gatech.edu/fce/ahri/>

²¹Microsoft Research EasyLiving homepage: <http://research.microsoft.com/easyliving/>

FormalInformalChildish.640190
☐ Formal.640191 ☐ Informal.640192 ☐ Childish.640193

Frequency.640150
☐ Never.510020 ☐ Sometimes.510021 ☐ Regularly.510022 ☐ Often.510023 ☐ Always.510024

NoiseLevel.640160
☐ Quiet.640161 ☐ Normal.640162 ☐ Noisy.640163 ☐ Loud.640164 ☐ ExtremelyLoud.640165

PoorGoodPerfect.640040
☐ Poor.640041 ☐ Acceptable.640042 ☐ Good.640043 ☐ VeryGood.640044 ☐ Perfect.640045

Figure 5.34: Special sets defined as data types, as shown in the *UbisOntologyBrowser*

aware homes can be found at [Meyer and Rakotonirainy, 2003]. More recent European smart home research projects are for example the *Philips HomeLab*²² in Eindhoven, the Netherlands, (figure 5.35(b) shows a prototypical display in a mirror), the “Haus der Gegenwart”²³ in Munich, Germany, see figure 5.36 and the “Intelligent House Duisburg Innovation Center - inHaus”²⁴ by Fraunhofer-Gesellschaft, see figure 5.37. The latter one includes apart from a residential home also a networked garden and a networked car. The basic idea is that not only all devices and components in the intelligent house communicate with each other, but they are also networked to the outside world, which allows distance monitoring and distance control.



Figure 5.35: (a) *AwareHome* by Georgia Tech, (b) *HomeLab* in Eindhoven by Philips

An early application of intelligent objects in the ubiquitous computing office scenario

²²Philips HomeLab: <http://www.research.philips.com/technologies/misc/homelab/>

²³Haus der Gegenwart homepage: <http://www.haus-der-gegenwart.de>

²⁴Fraunhofer-Gesellschaft inHaus homepage: <http://www.inhaus-duisburg.de/>

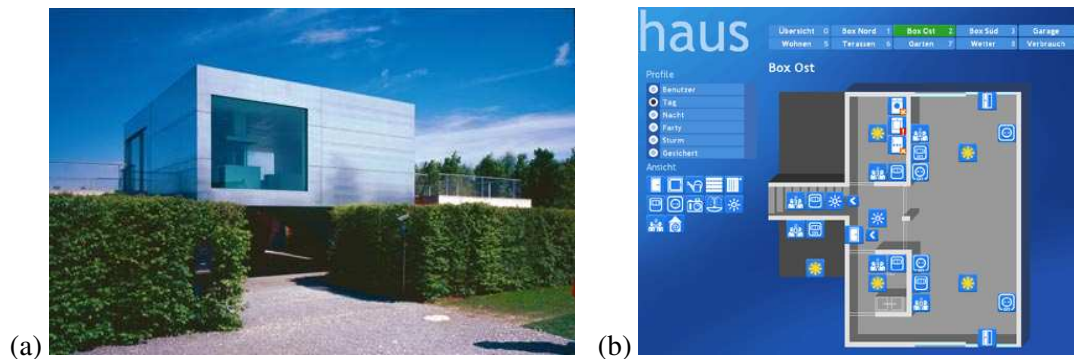


Figure 5.36: “Haus der Gegenwart” in Munich with its graphical user interface by Microsoft

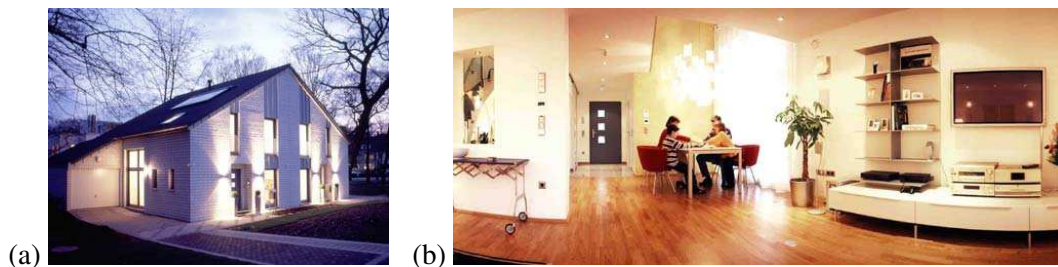


Figure 5.37: “inHaus” by Fraunhofer-Gesellschaft with its intelligent living room

was the MediaCup²⁵ by the University of Karlsruhe, see figure 5.38(b). It is an ordinary coffee cup augmented with sensing, processing and communication capabilities that are integrated in the cup’s bottom, to collect and communicate general context information in a given environment. The electronics in the cup senses how the cup is used. For example if someone drinks out of the cup, if someone plays with it and the temperature of the cup. This information is then communicated to other computerized objects in the room like a coffee machine or a Web server. Programs running on these objects use this data to support the human using these systems. Small Meeting of figure 5.38(a) describes the detection process of small meetings, for example if several cups of hot coffee realize their coexistence in a meeting room and deduce themselves that a meeting takes place, see [Gellersen et al., 1999].

The inference ontology, which is still under construction²⁶, is especially introduced to collect all inferences the lead to intelligent interaction behavior in the research of ubiquitous computing. Remind me there and Remind me then represent virtual notes that can be posted within the whole UbiWorld as *Digital Graffiti*. A real world example of this technology is for example realized in the SPELLBINDER²⁷ project. These there & then rules are used to demonstrate basic proactive behavior within this virtual world. Figure 5.39 shows

²⁵MediaCup homepage: <http://mediacup.teco.edu/>

²⁶The current status can be found at <http://www.ubisworld.org>

²⁷SPELLBINDER homepage: <http://spellbinder.inf.ed.ac.uk/>

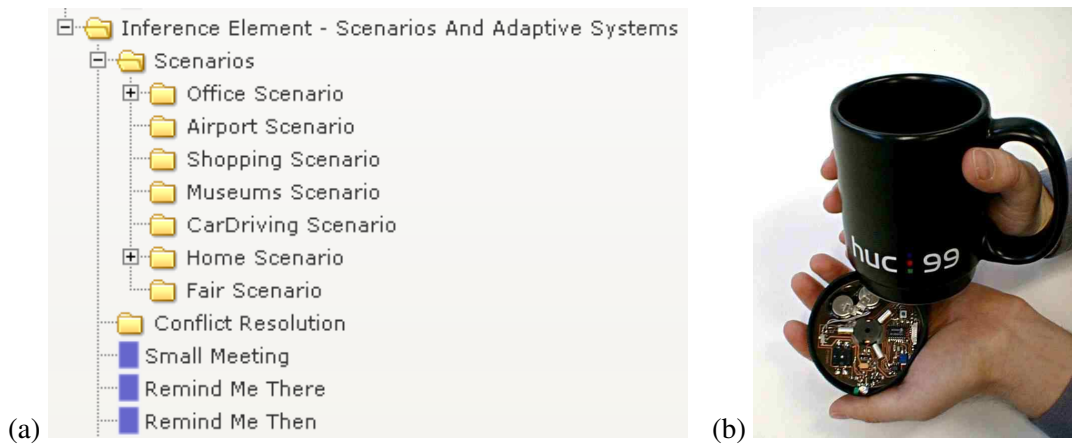


Figure 5.38: (a) Collection of inference elements, (b) MediaCup by the University of Karlsruhe

the screen-shot of a digital graffiti trigger editor. The currently edited rule says that *everytime, the person Jörg enters the restaurant Schlemmer Eule, he will be reminded with a sentence that a certain dish is known to be very good there.*

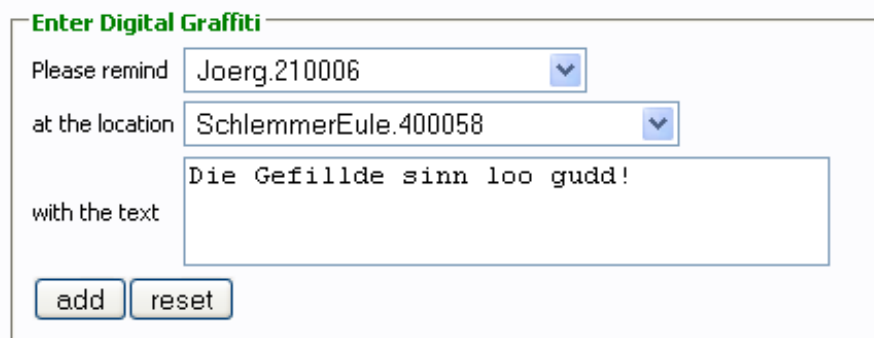


Figure 5.39: Screen-shot of the digital graffiti trigger editor in UbiWorld

To summarize, *UbiWorld* has been developed under the impression of the real world with ubiquitous computing and user modeling. All elements are defined in six open ontologies: the physical ontology, the spatial ontology, the temporal ontology, the activity ontology, the situation ontology (which contains the general user model ontology GUMO), and the inference ontology. Altogether, they form the UBISWORLD ONTOLOGY. Several new conceptual models have been integrated into this ontology. All concepts have been implemented and tested, while several instances of UbiWorld servers are currently running, either locally or on the internet²⁸.

²⁸UbiWorld can be tested at: <http://www.ubisworld.org/>